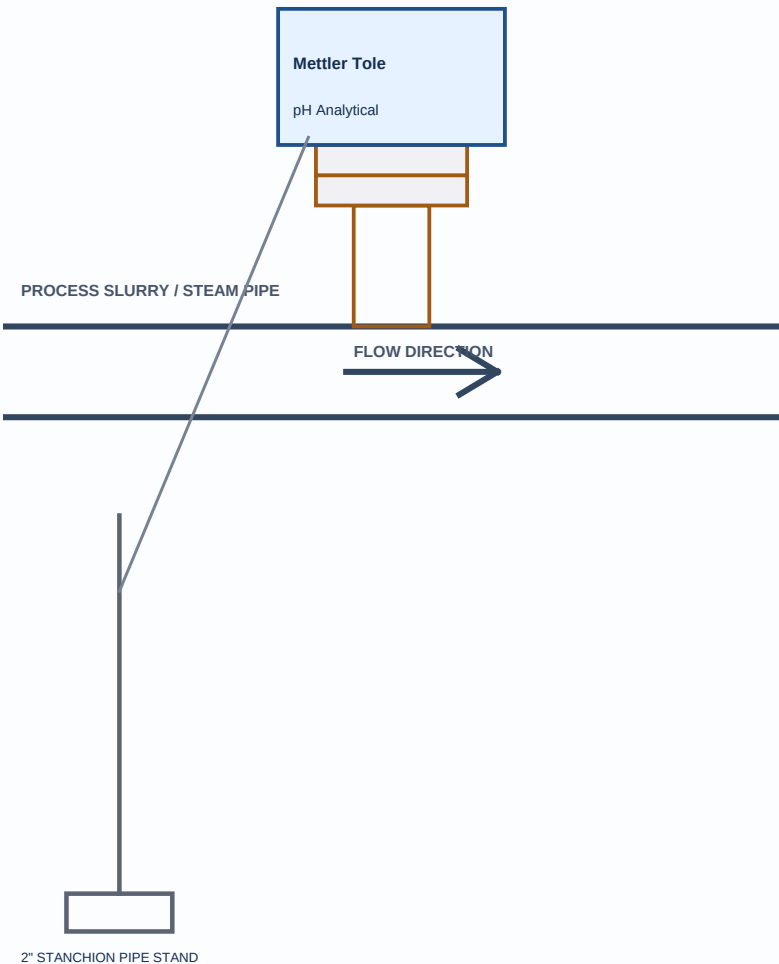


TEMPLATE TYP-03: PH ANALYTICAL LOOP

TYPICAL TAGS: AIT-40101 / AIT-40201

SERVICE: Cooker Slurry & Flash Tank pH Neutralization | PLC I/O: 1756-IF16 (AI pH) & 1756-IB32 (DI Alarms)

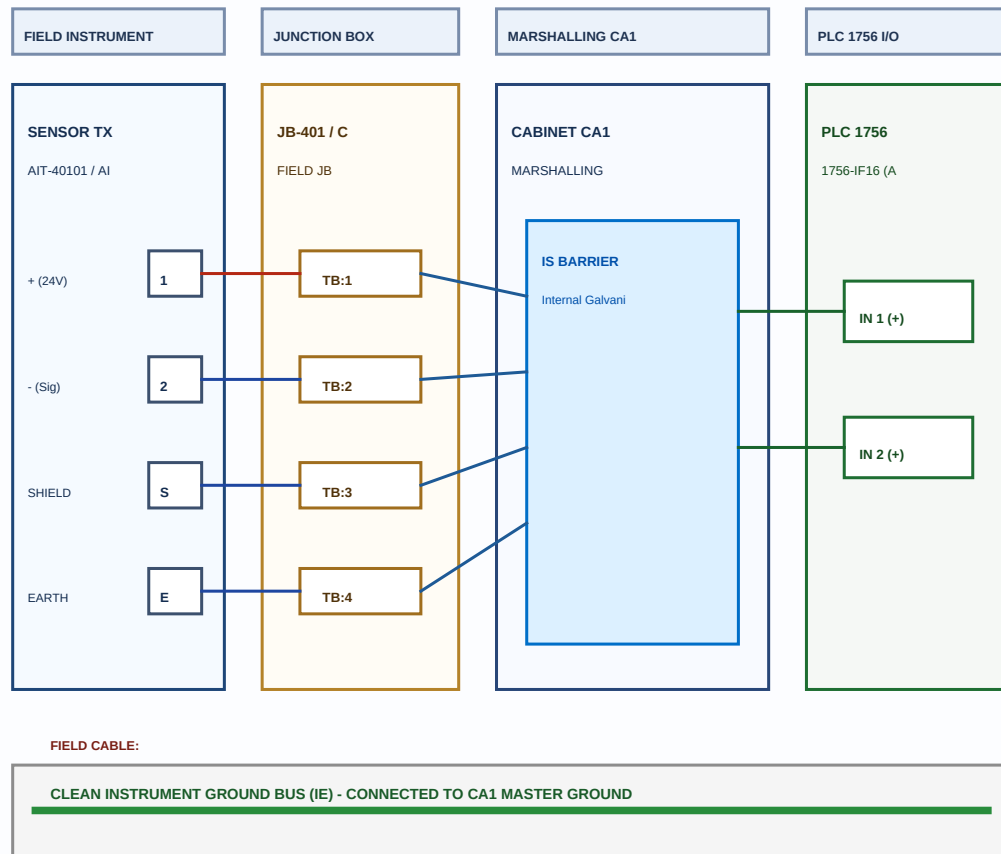
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN25 Weld-in Socket (Incline..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP65 Polycarbonate..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6/T4..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	AI: 0-14 pH (4-20mA HART) / AI: Temp
Power Supply:	24V DC / 10W (Optional 100-240V AC)
Field Cable Type:	VP6 High-Z Cable to Tx + 2Px1.5mm² Shi..
Destination JB:	JB-401 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Internal Galvanic Isolation (4-Wire Tx)
PLC Module:	1756-IF16 (AI pH) & 1756-IB32 (DI Alar..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: METTLER TOLEDO

MODEL: M300G2 4-Wire Transmitter + InPro3250i + InTrac 777P

SERVICE: Cooker Slurry & Flash Tank pH Neutralization

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	pH Multi-Parameter Transmitter Mettler Toledo M300 4-Wire		1 EA
2	Intelligent pH Sensor InPro325 Pre-pressurized Liquid Gel		1 EA
3	Retractable Housing InTrac 777 SS316L / Hastelloy C w/ Flange		1 EA
4	Weld-in Socket 40mm / DN25 (15 SS316L Sloped Socket		1 EA
5	2" Pipe Mounting Kit for M300G SS304 Bracket & U-Bolts		1 SET

ENGINEERING INSTALLATION NOTES

- Weld socket must be installed inclined at min. 15° above horizontal to prevent air bubble entrapment.
- Use retractable housing safety interlock valve before extracting probe during process operation.
- Ground transmitter chassis directly to plant PE bar with min. 4mm² copper grounding strap.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTpH ANALYTICAL MEASUREMENT LOOP HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDWG NO: KAL-JC-TYP-003
REV: 3.6 DATE: 2026-09-08PROJECT NO:
R5THS00138-JCSHEET NO:
05 / 25